



1495

UNIVERSITY OF  
**ABERDEEN**

CELEBRATING  
**525 YEARS**  
1495 – 2020

ABERDEEN 2040

# Image Mining – 1

Data Mining & Visualisation  
Lecture 29

2025



# Today...

- Digital Images
- Image Enhancement and Preprocessing

# Image Mining

As smartphones, digital cameras, CCTV, and embedded image sensors become increasingly ubiquitous, the volume of image data being captured and stored continues to grow rapidly.

As a result, the tools, techniques, and academic disciplines concerning the analysis of image data are also continuing to develop and evolve.

# Image Mining

We're going to discuss some of the key aspects of how digital images can be represented, processed and analysed.

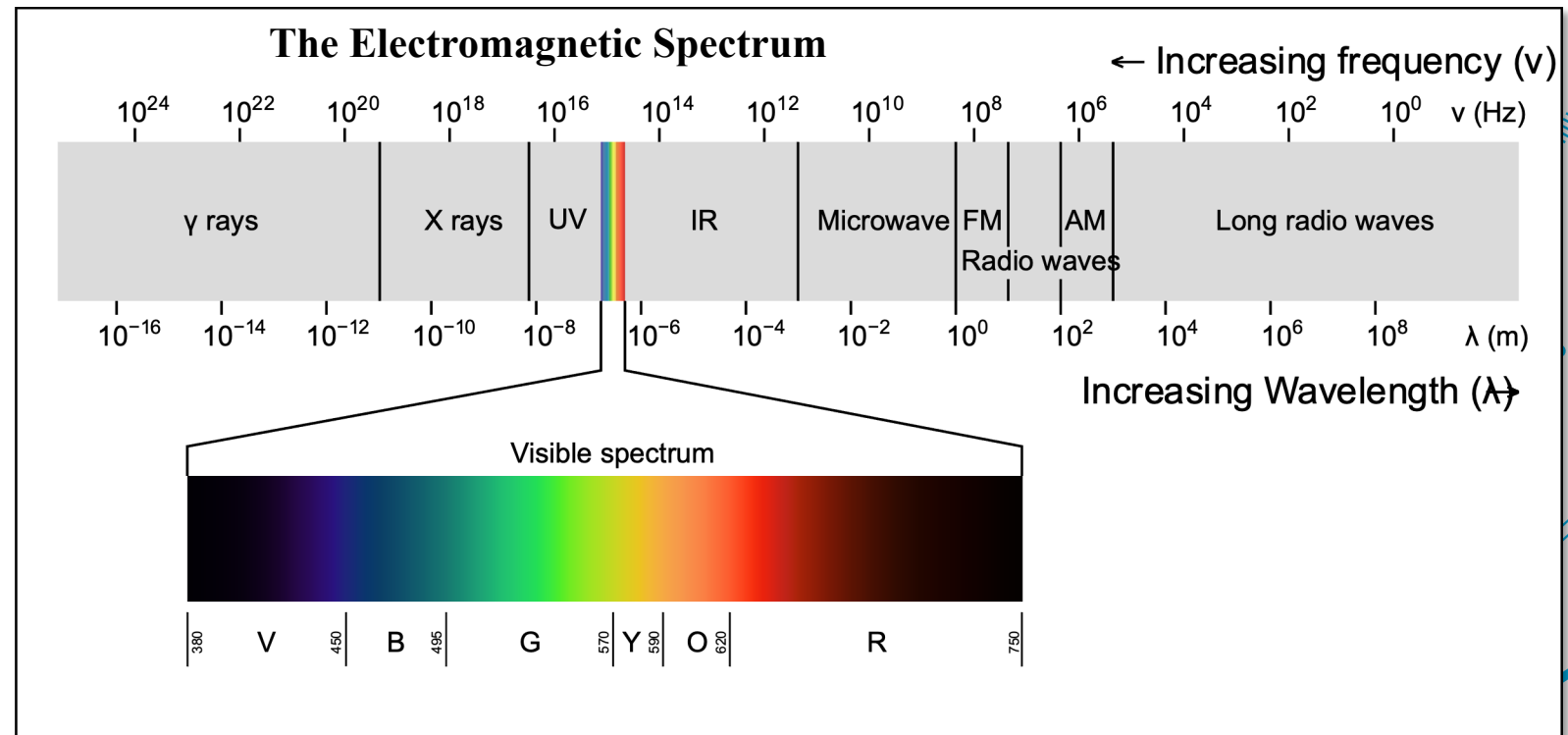
To start off, it's useful to have a basic understanding of how light, the human perception system, and digital cameras work.

# Digital Images



# The Electromagnetic Spectrum

What we think of as (visible) light is a small part of a spectrum of electromagnetic energy, containing radio waves, microwaves, infrared and ultraviolet light, and X and gamma rays.



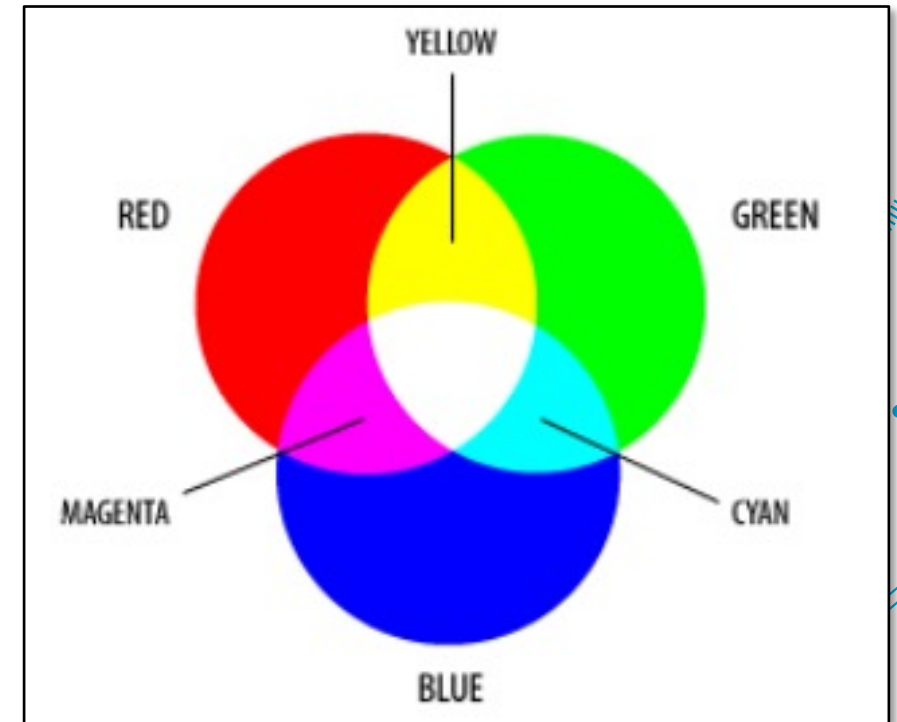
# The Electromagnetic Spectrum

Within that visible spectrum, the wavelength determines how we perceive it.

We see:

- ~ 700 nm wavelength light as red
- ~ 550 nm wavelength light as green
- ~ 450 nm wavelength light as blue

Different combinations of these wavelengths result in different colours (e.g. red + green = yellow).



# Light Reflection

When we 'see' an object, what we are actually seeing is light reflecting off of that object.

The object absorbs certain wavelengths of light and reflects other wavelengths.

Those reflected wavelengths define what we see and what colour(s) they appear to be.





# Human Perception System

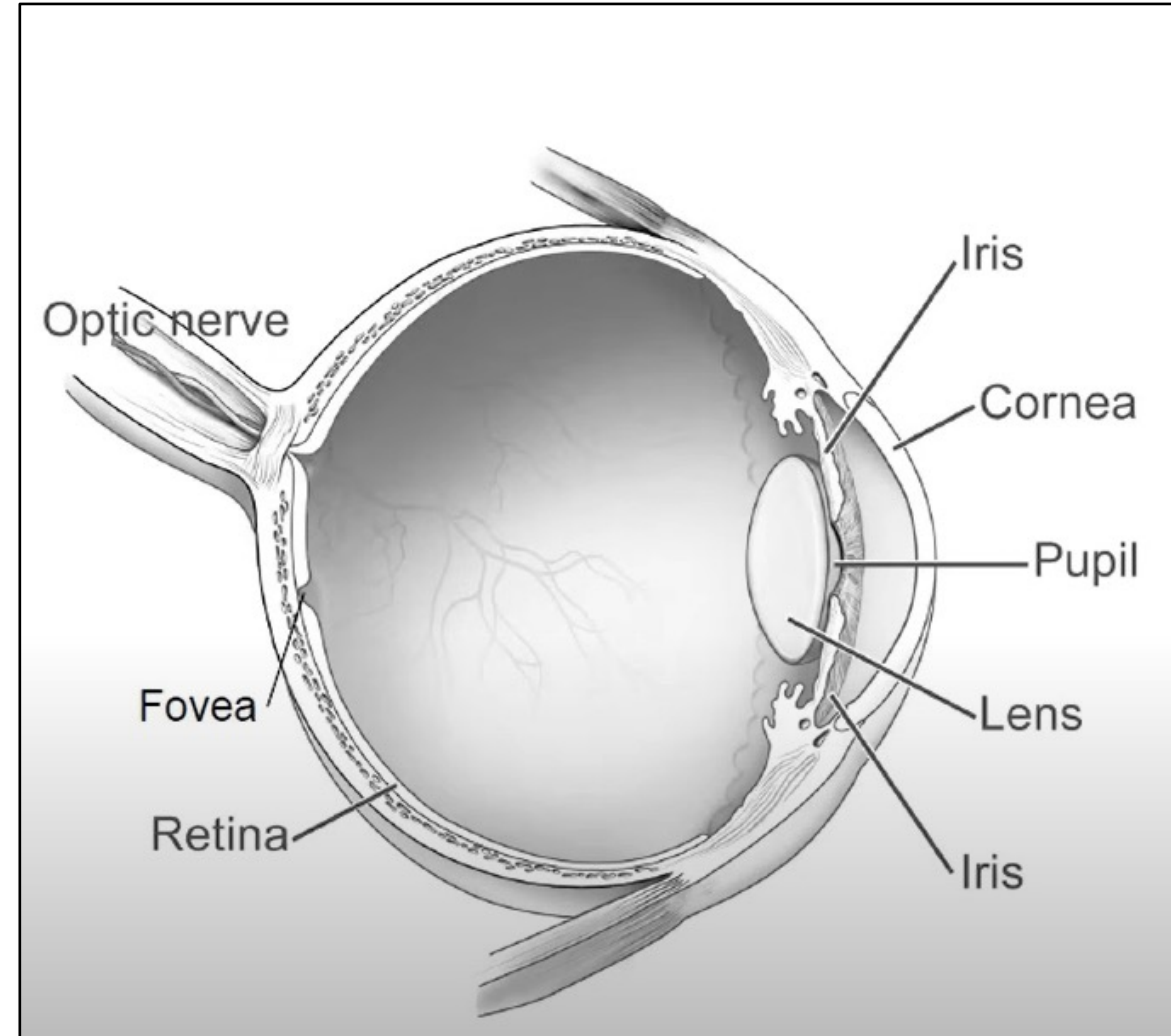
Light enters the **pupil**.

The **iris** contracts and expands, controlling how much light enters the eye.

The **lens** focuses the light onto the retina, where light sensitive cells are.

The **fovea** is where the highest concentration of light sensitive cells are.

The **optic nerve** carries all of these signals to the brain, for processing.



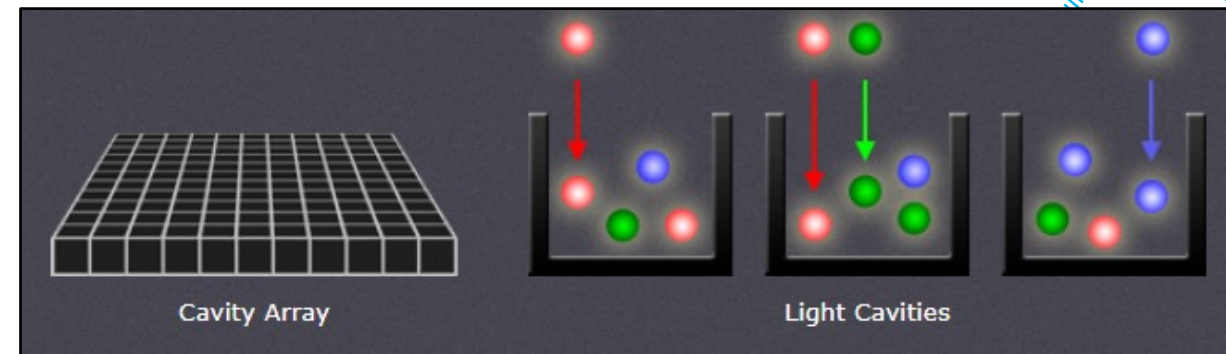
# Digital Cameras

The way that digital cameras work is broadly inspired (to a degree) by the human perception system.

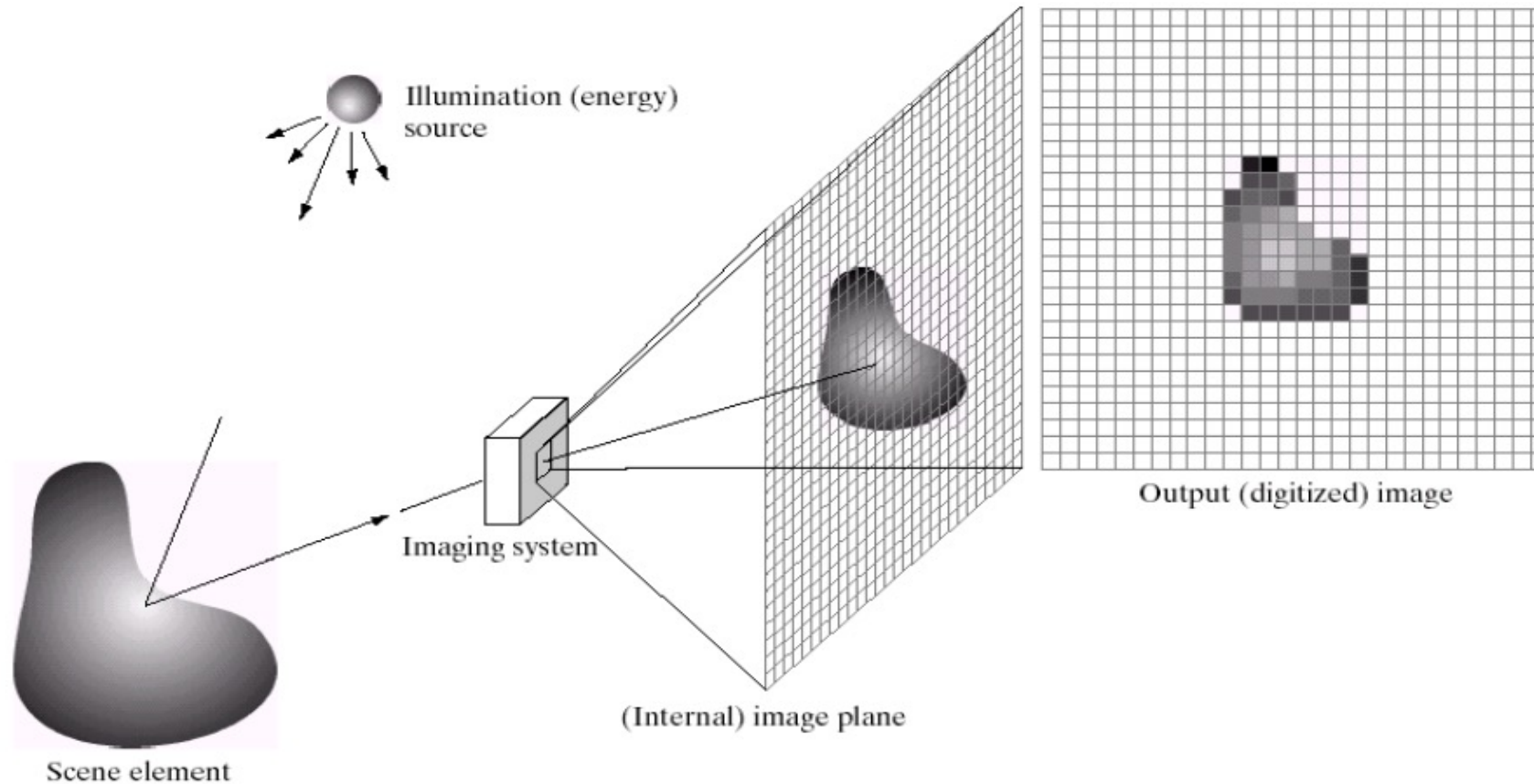
They have an array of light sensitive sensors (called photosites).

The camera's shutter opens, exposing this array to a stream of light particles.

When the shutter closes, the camera measures the strength of electric signal for each photosite, corresponding to the amount of light that it was exposed to.



# Image Acquisition and Representation



An example of the digital image acquisition process. (a) Energy source, (b) An element of a scene. (c) Imaging system. (d) projection of the scene onto the image plane. (e) digitalized image.

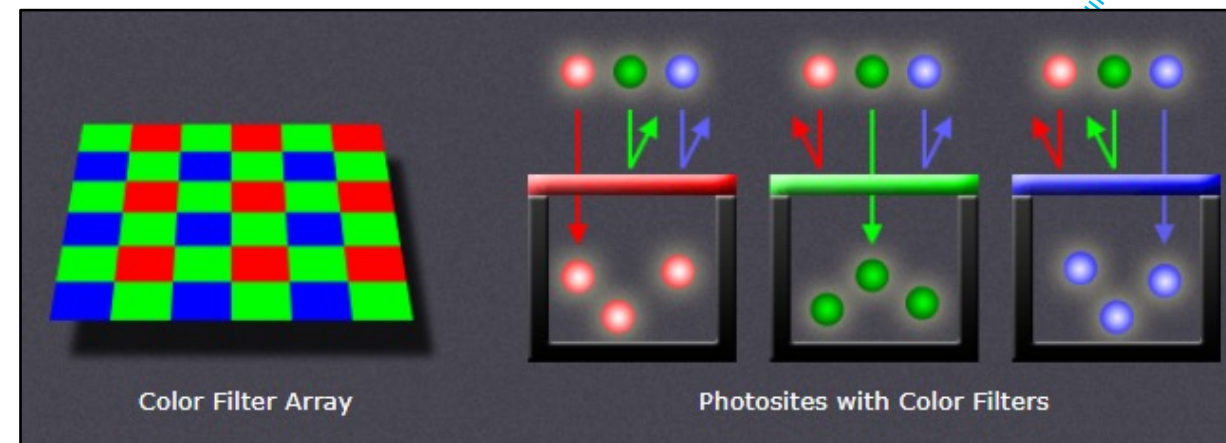
# Digital Cameras

More accurately, each of these photosites will typically have a colour filter applied.

These filters only allow certain light wavelengths to pass, so that only red light is absorbed by a photosite with a red filter, green light by a green filter, etc.

These colour filtered photosites are typically arranged in a series of repeating 2x2 patterns, containing:

- 1x red filtered photosite,
- 1x blue filtered photosite, and
- 2x green filtered photosites.

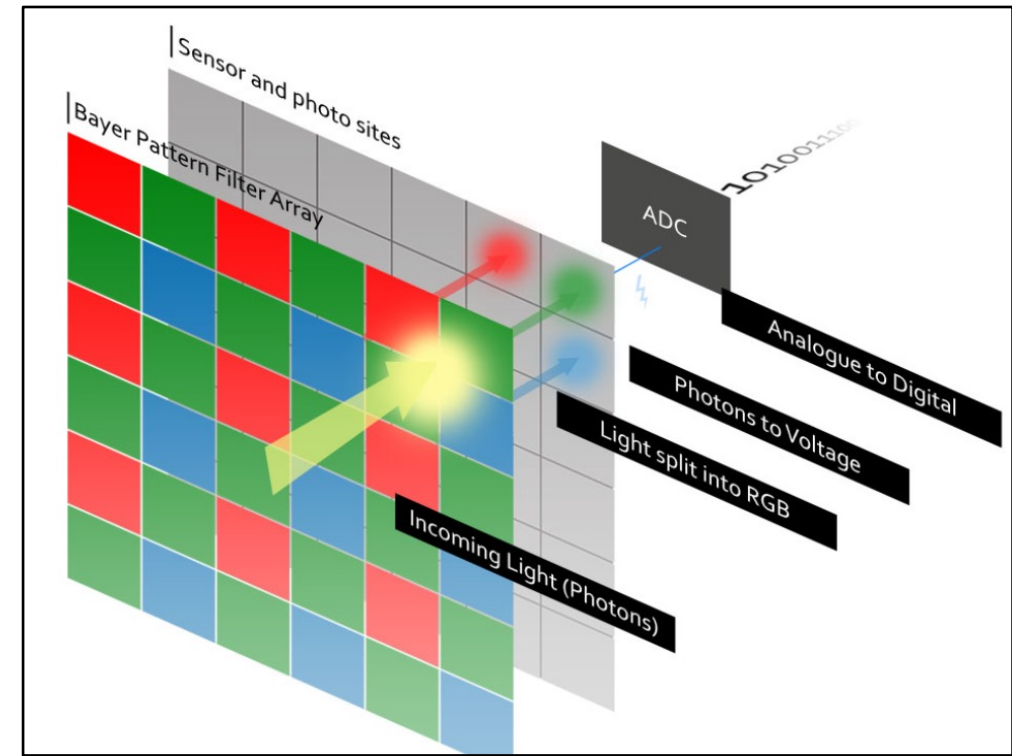


# Digital Cameras

This 2x2 pattern is called a Bayer array.

After the camera's shutter is closed, each photosite's electric signal is converted into an 8-bit digital value.

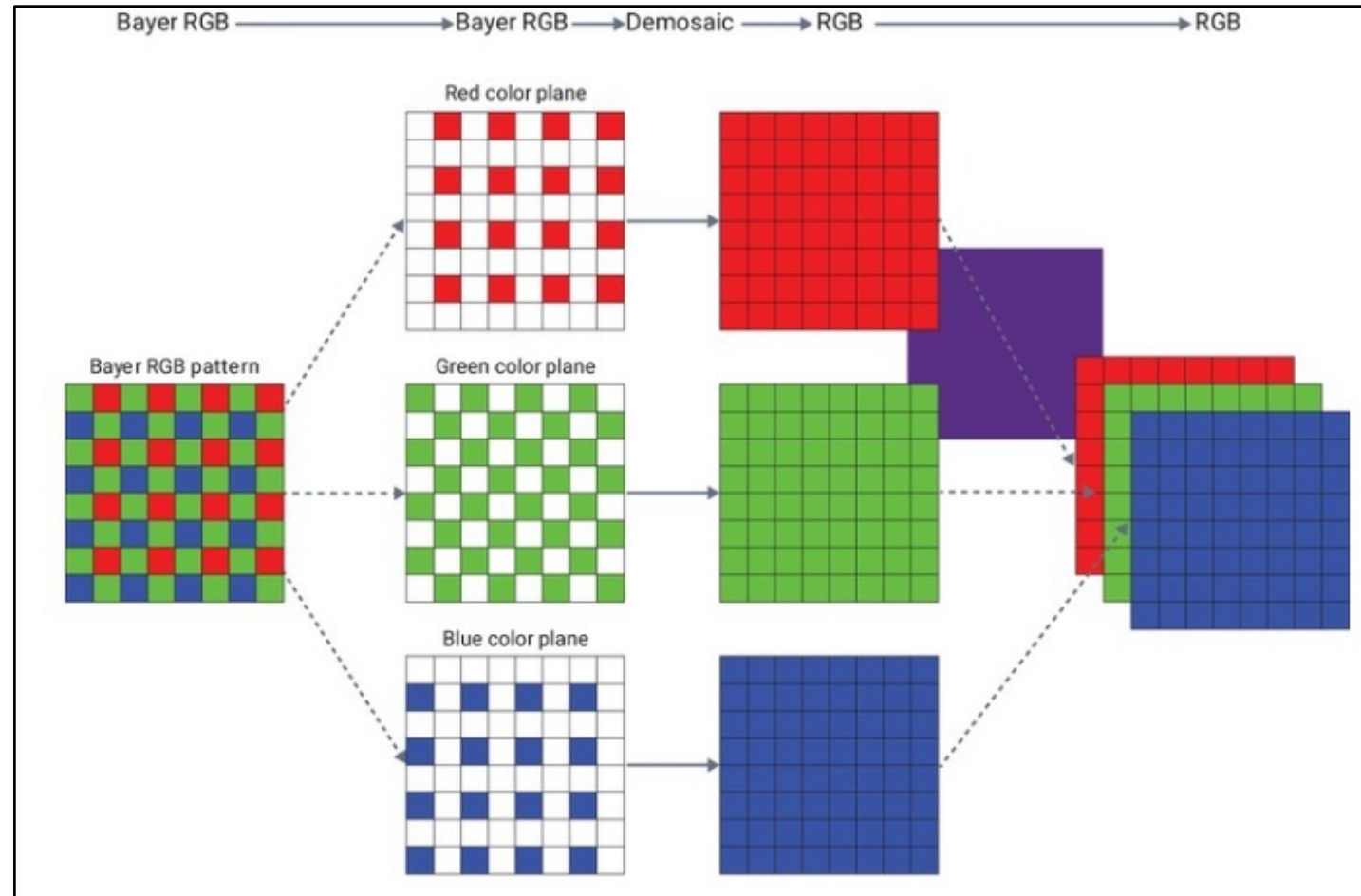
These individual 8-bit values are then processed together to generate RGB pixel values.



# Demosaicing

The digital values for each of these 2x2 patterns are used to reconstruct full RGB values for each pixel, using interpolation.

This process is known as **demosaicing**.

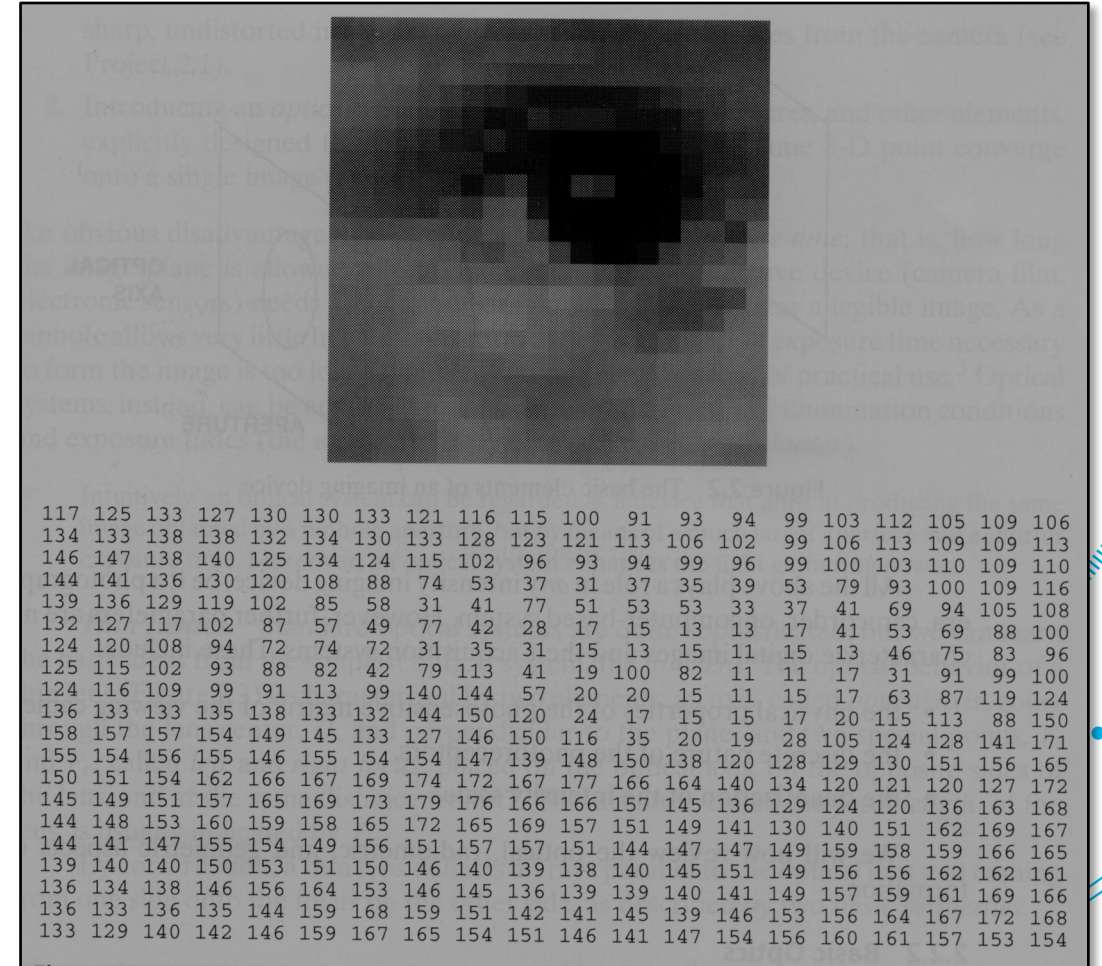




# Image Representation

Digital images are represented by a 2-dimensional grid of pixel values:  
 $P[x, y]$

The overall size of the image is  $x*y$  pixels.



# Image types

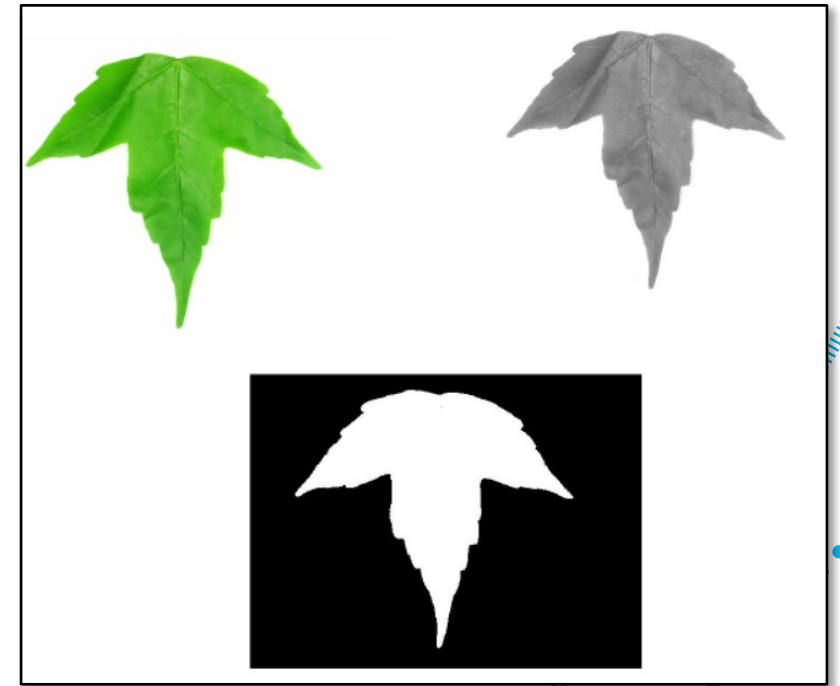
Pixels can be represented in different ways, which will change the 'type' of image.

This includes:

**RGB images:** Three 8-bit channels; R, G, and B

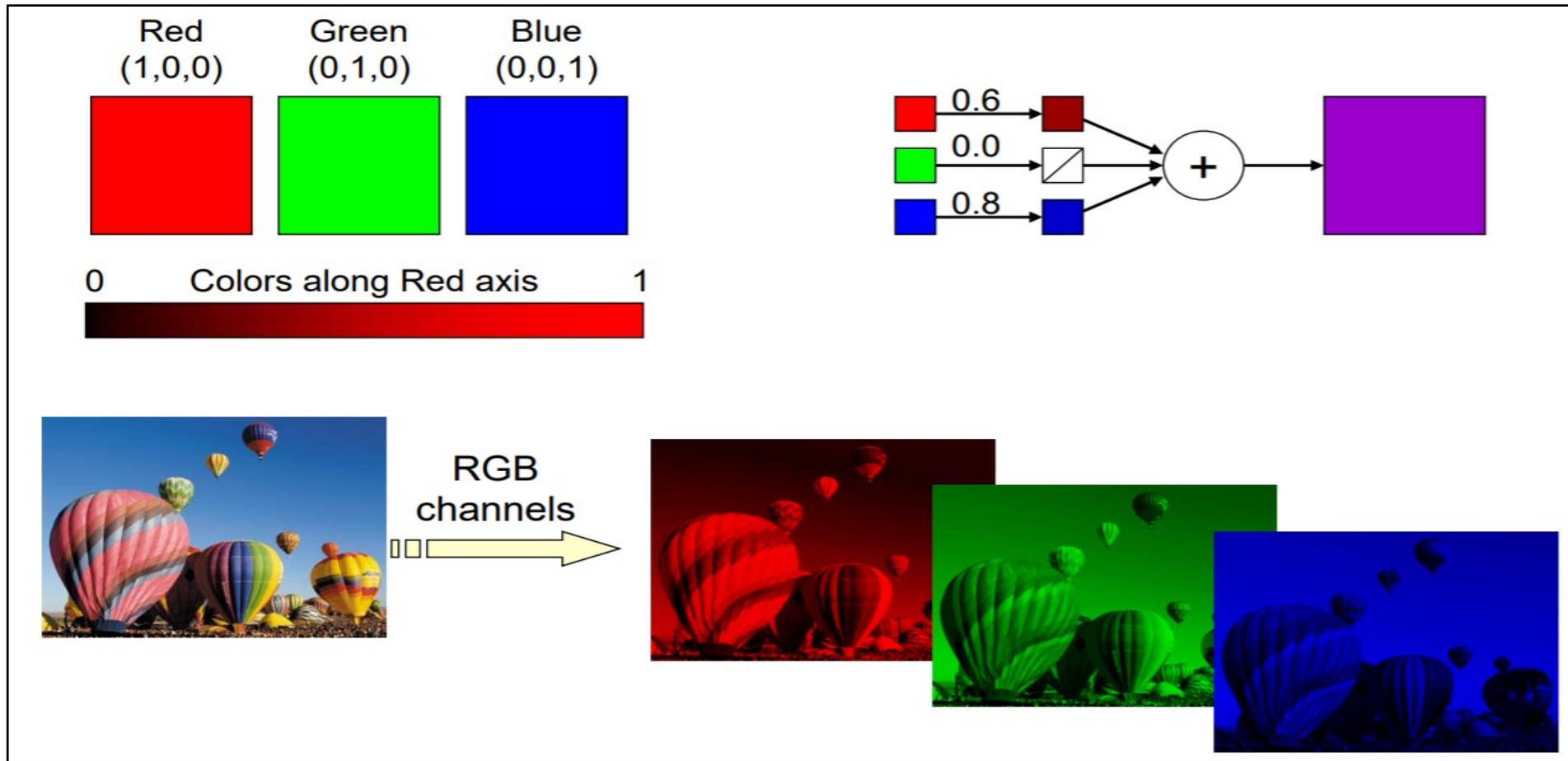
**Grayscale images:** One 8-bit channel

**Binary images:** One 1-bit channel (true or false)





# RGB Images



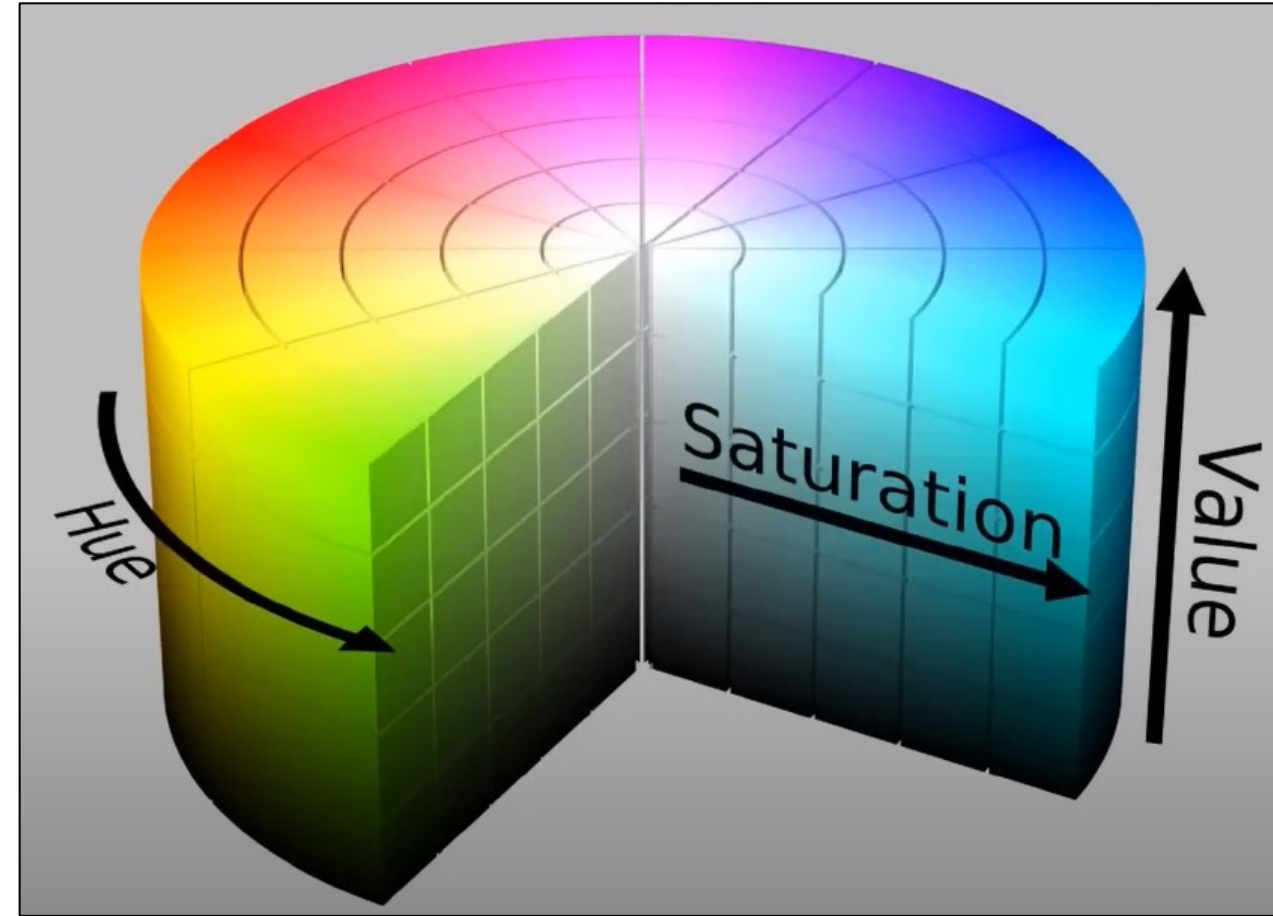
# Colour Terms

RGB pixels can be described in terms of their:

**Intensity / brightness / value:** i.e. how much light the pixel contains

**Hue:** the dominant colour

**Saturation:** the purity or vividness of the colour



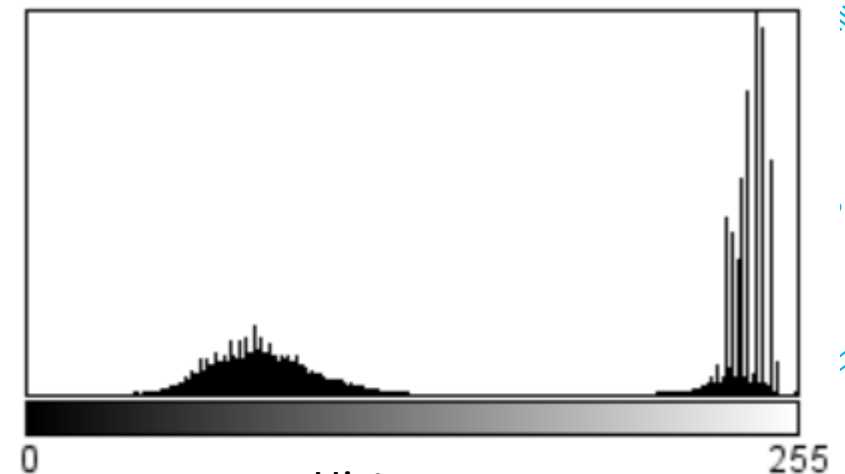
# Image Histogram

Image histograms show us the frequency distribution of intensity values within the image.

They plot the number of pixels corresponding to each pixel value.

By looking at the histogram, we can judge the entire tonal distribution of the original image at a glance.

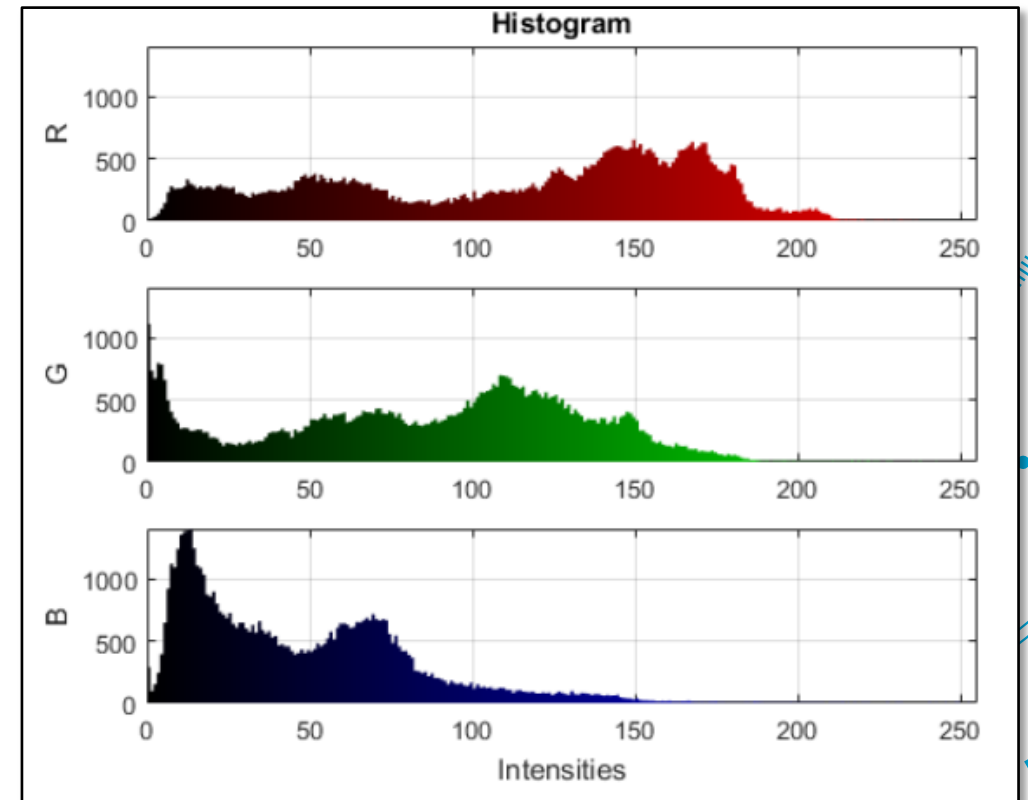
Grayscale image



Histogram

# Image Histogram

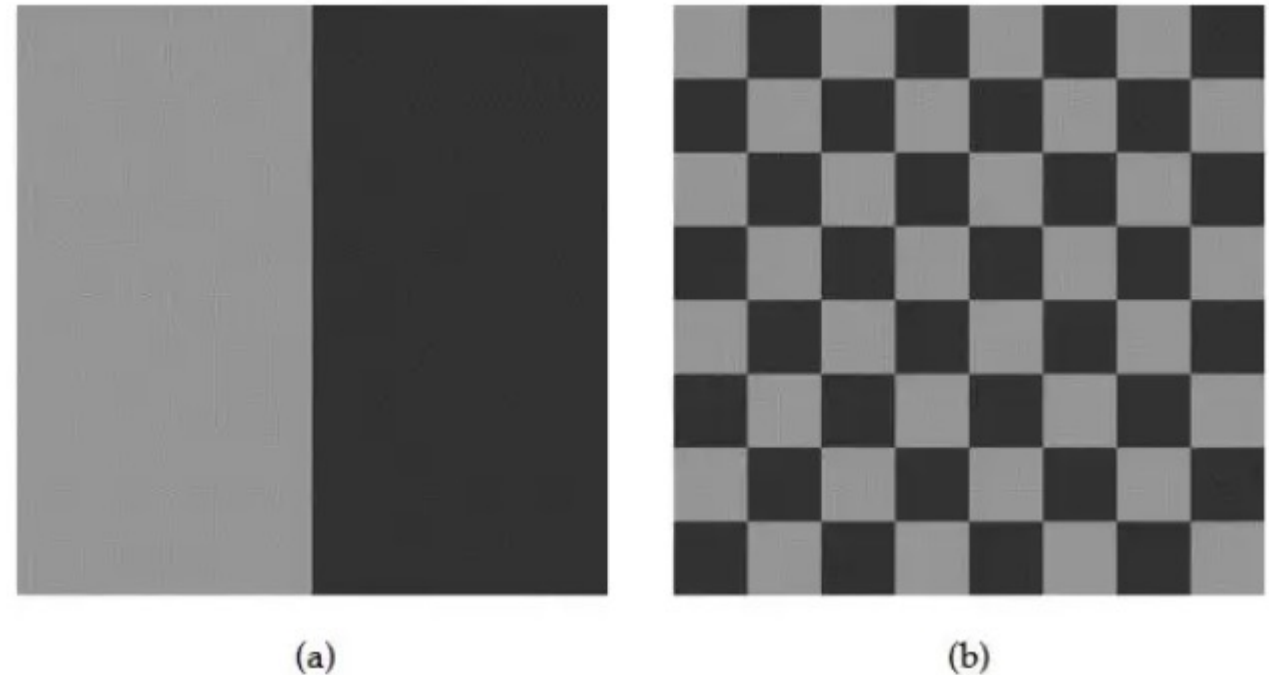
For RGB images, the idea is the same, but we can also have a separate histogram representing each colour channel.



# Histogram

Importantly, histograms only provide statistical information about the distribution of colour values.

They do not retain any of the *spatial* information about the picture itself.



Two different pictures, a. and b., provide the same histogram due to the equal number of the intensities



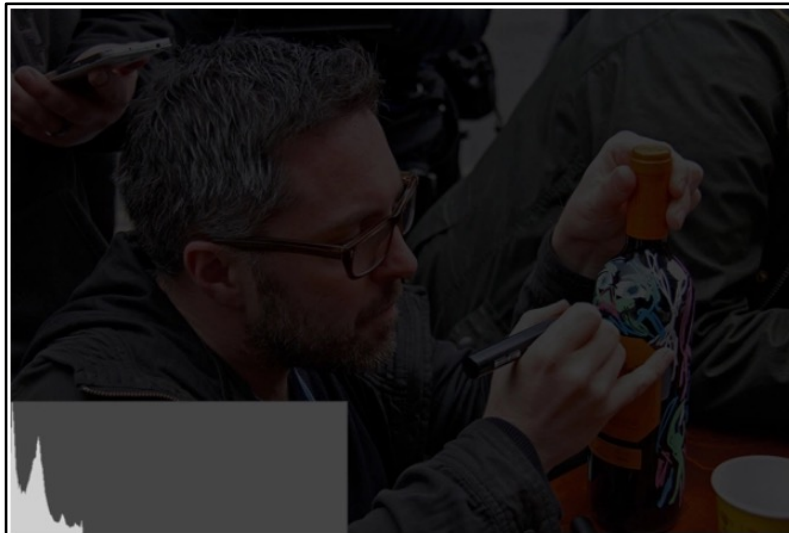
# Image Quality

However, they can give us information about the image itself (i.e. whether it is under- or over-exposed).

Original



Too dark (under-exposed)



Too light (over-exposed)



# Image Quality

What we're often looking for in an image is having a balanced histogram.

I.e., showing a bell-shaped curve.





# Image Enhancement and Preprocessing



# Image Enhancement (Point Operation)

**Point Operation** is the term given to modifying the pixel values of an image, without changing in the size, geometry or local structure of that image.

The new value of any given pixel depends only on that pixel's previous value.

# Image Enhancement (Point Operation)

Point operations might include:

- Modifying the brightness or contrast of an image
- Clamping (restricting the range) of pixel values
- Inverting the image
- 'Thresholding' the image

# Image Enhancement – Brightness

For example, if we wished to adjust the brightness of our image, we could simply add (or subtract) some fixed intensity value to each pixel.

Pixel values would then be ‘clamped’, to restrict the pixel values to the full intensity range (i.e. 0-255 for 8-bit images).

# Image Enhancement – Contrast Stretching

To adjust the contrast of an image, we can map the lowest and highest intensity values of a particular image to the *min* and *max* values of the full intensity range.

$$f_{ac}(a) = a_{\min} + (a - a_{\text{low}}) \cdot \frac{a_{\max} - a_{\min}}{a_{\text{high}} - a_{\text{low}}}$$

For an 8 bit image:

$$f_{ac}(a) = (a - a_{\text{low}}) \cdot \frac{255}{a_{\text{high}} - a_{\text{low}}}$$

Where:

$a$  is a pixel intensity value,

$a_{\text{low}}$  is the **lowest** intensity value of the image, and

$a_{\text{high}}$  is the **highest** intensity value of the image

# Image Enhancement – Inversion

Inversion is the process of inverting an image, such that pixel values become 'flipped'.

Pixels with a high intensity value become low, and pixels with a low intensity value become high.





# Image Enhancement – Inversion

For each pixel, we simply subtract the pixel's intensity value from the max intensity value.

I.e. for 8-bit images:  
 $\text{inversion}(a) = 255 - a$

Original Image

|     |     |     |
|-----|-----|-----|
| 0   | 255 | 127 |
| 10  | 250 | 50  |
| 200 | 100 | 50  |

8-bit & (3 X 3) Image

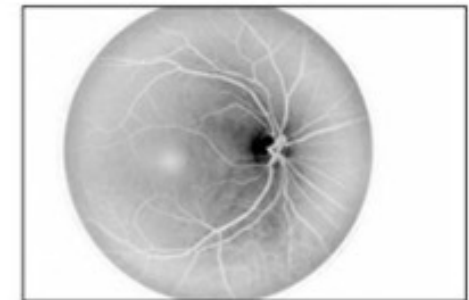
Negative Image

|     |     |     |
|-----|-----|-----|
| 255 | 0   | 128 |
| 245 | 5   | 205 |
| 55  | 155 | 205 |

Original Image



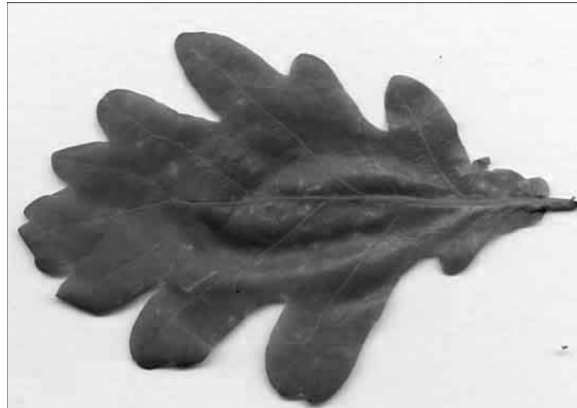
Negative Image



# Image Enhancement – Thresholding

Thresholding is the process of separating the foreground pixels from the background pixels.

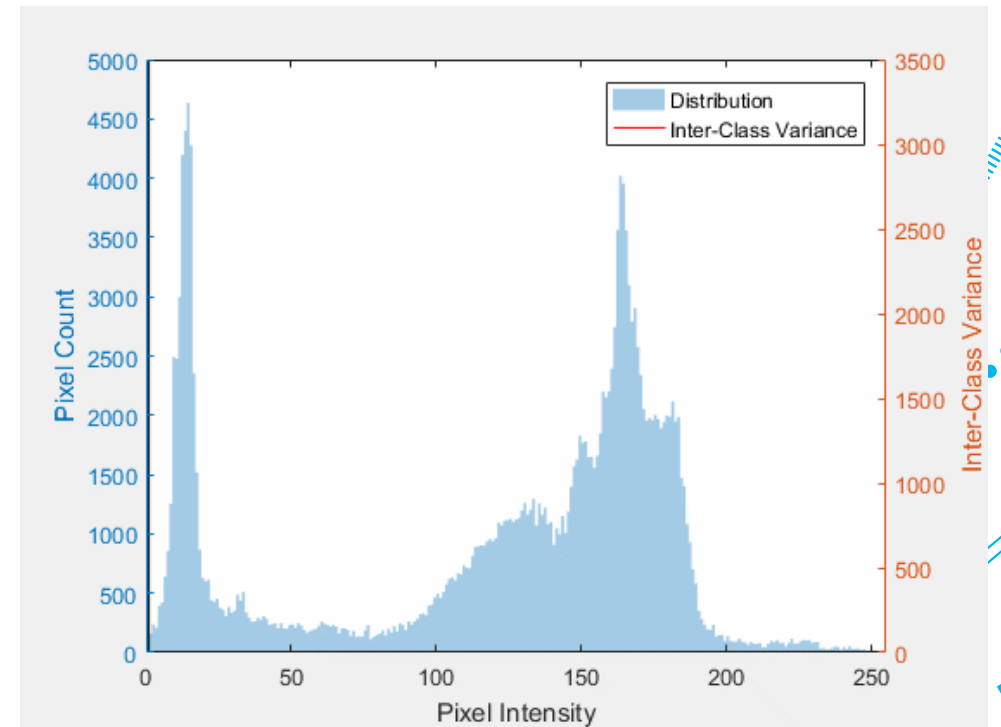
It results in the conversion of grayscale or RGB images into binary images.



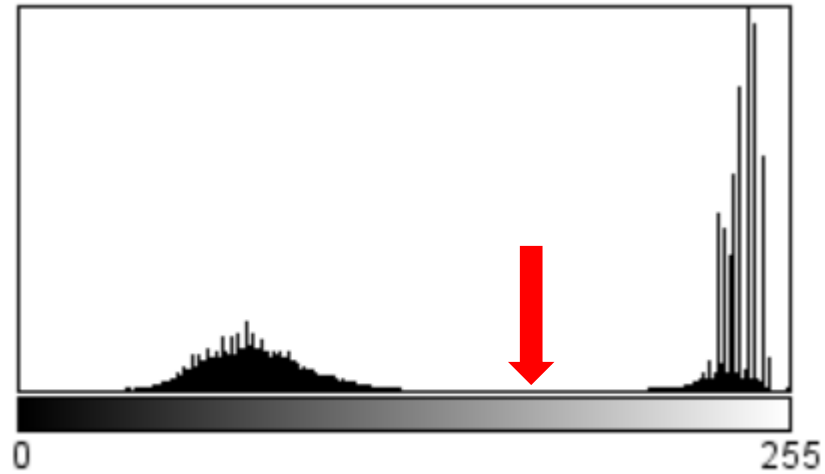
# Otsu Thresholding

One example is '**Otsu Thresholding**', which is a variance-based thresholding technique.

It finds the threshold value which **maximises the inter-class variance** between the foreground and background pixels.



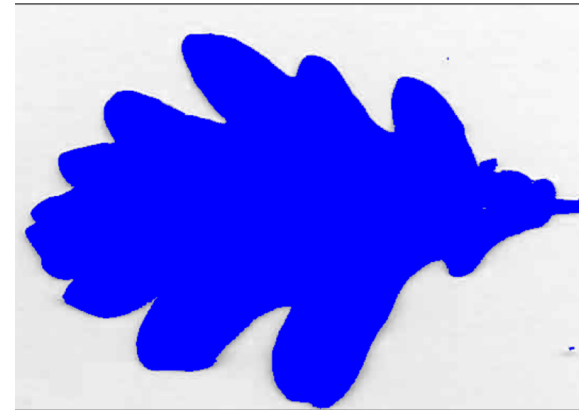
# Otsu Thresholding



RGB image  
(three bands)

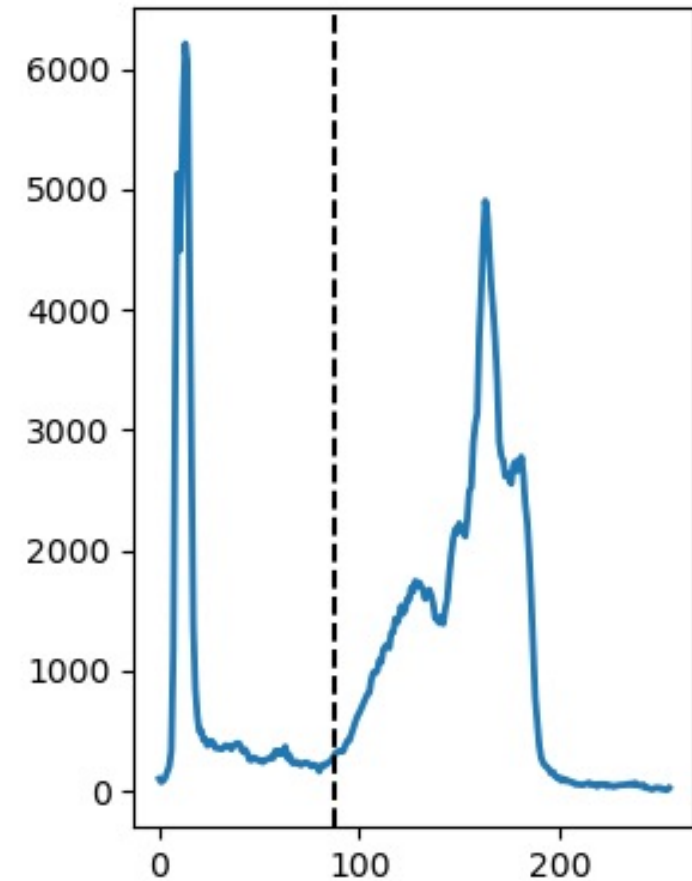


Grayscale image  
(one band)



binary image (0  
and 1)

# Otsu Thresholding



# Adaptive Thresholding

Adaptive Thresholding is another thresholding technique, which works better on images with varying brightness levels.

This approach allows us to get **different threshold values for different regions** of the image.





# Mathematical Morphology



# Mathematical Morphology

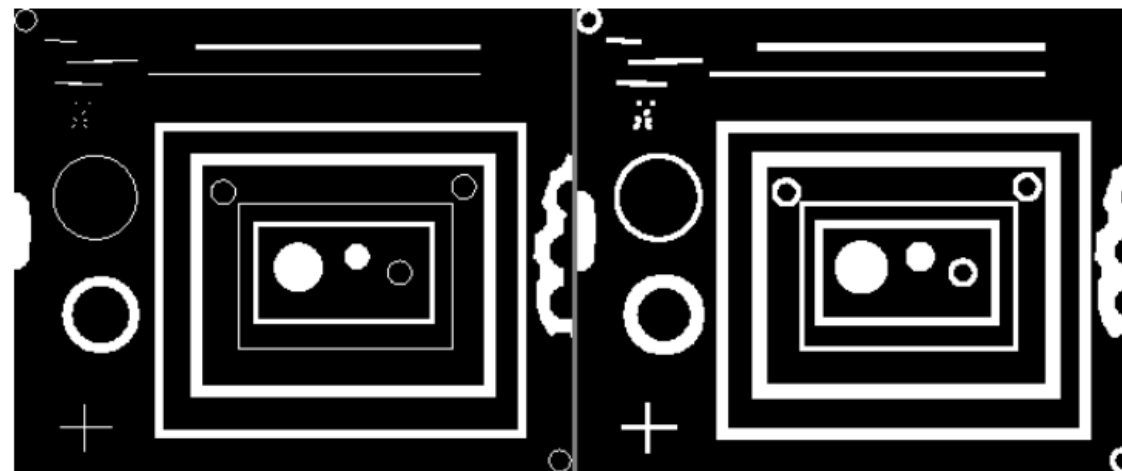
Mathematical morphology is a way of processing the structure and shape of objects in digital images, particularly binary and grayscale images.

It can be used for shape extraction, boundary detection, noise removal, and image segmentation.

# Mathematical Morphology

Binary mathematical morphology has two basic operations:

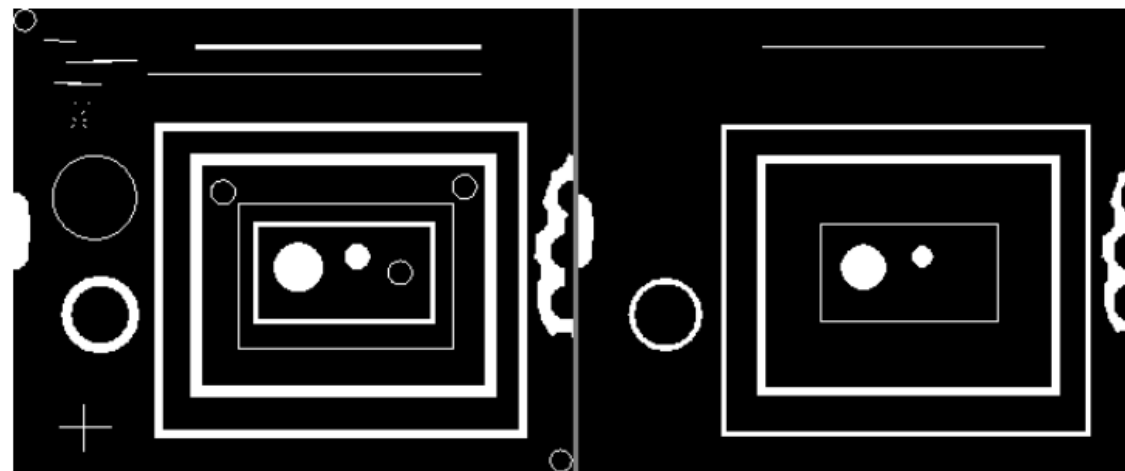
**Dilation:** making objects more visible and fills in small holes in objects. Lines appear thicker, and filled shapes appear larger.



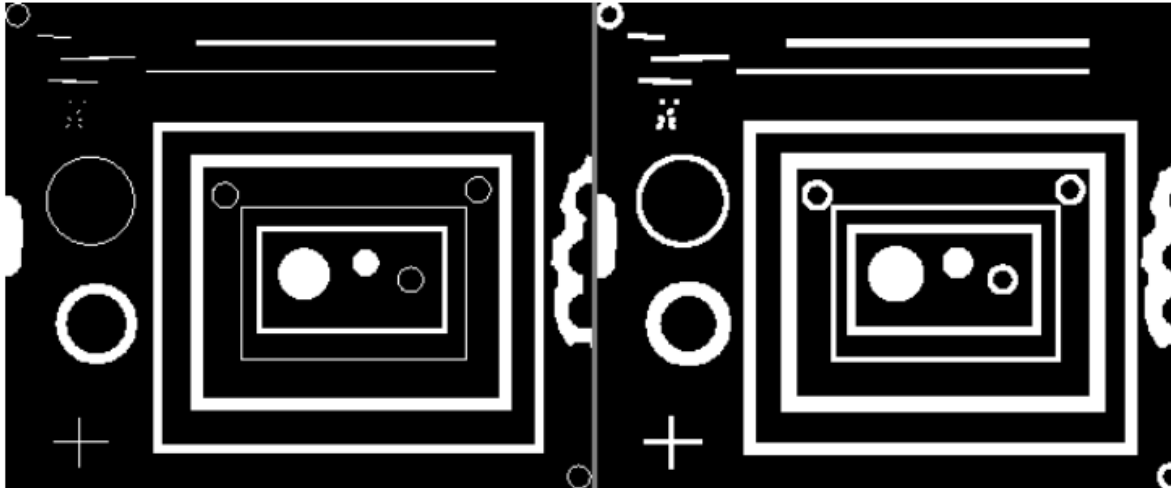
# Mathematical Morphology

Binary mathematical morphology has two basic operations:

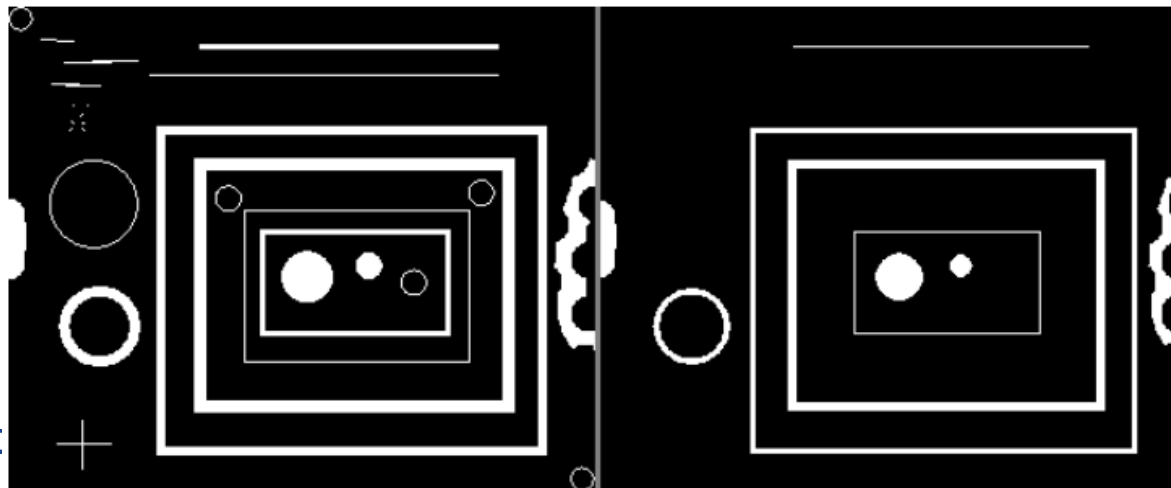
**Erosion:** Removing floating pixels and thin lines so that only substantive objects remain. Remaining lines appear thinner and shapes appear smaller.



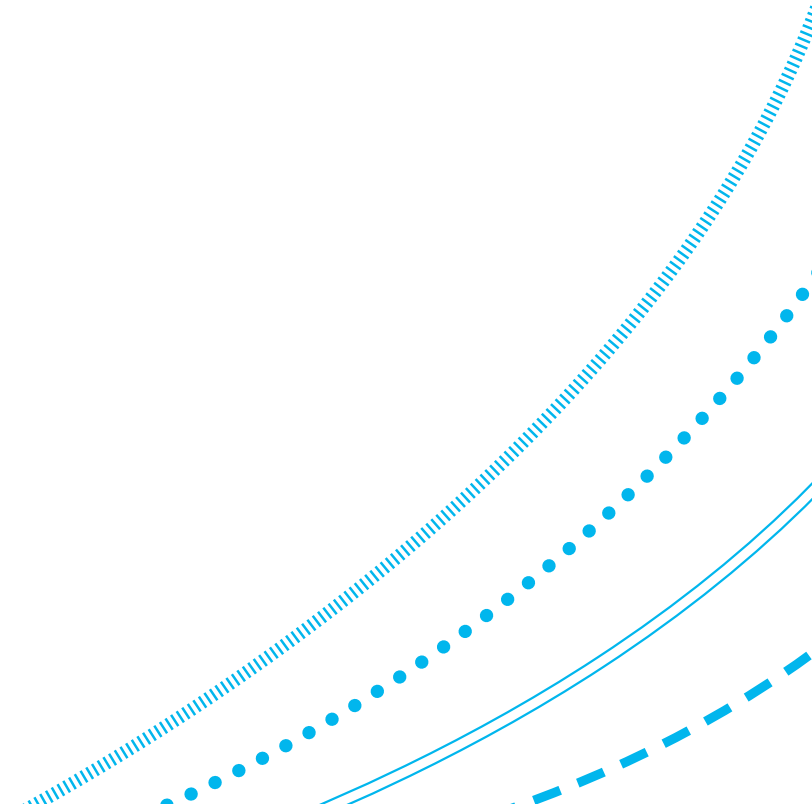
# Mathematical Morphology



**Dilation**



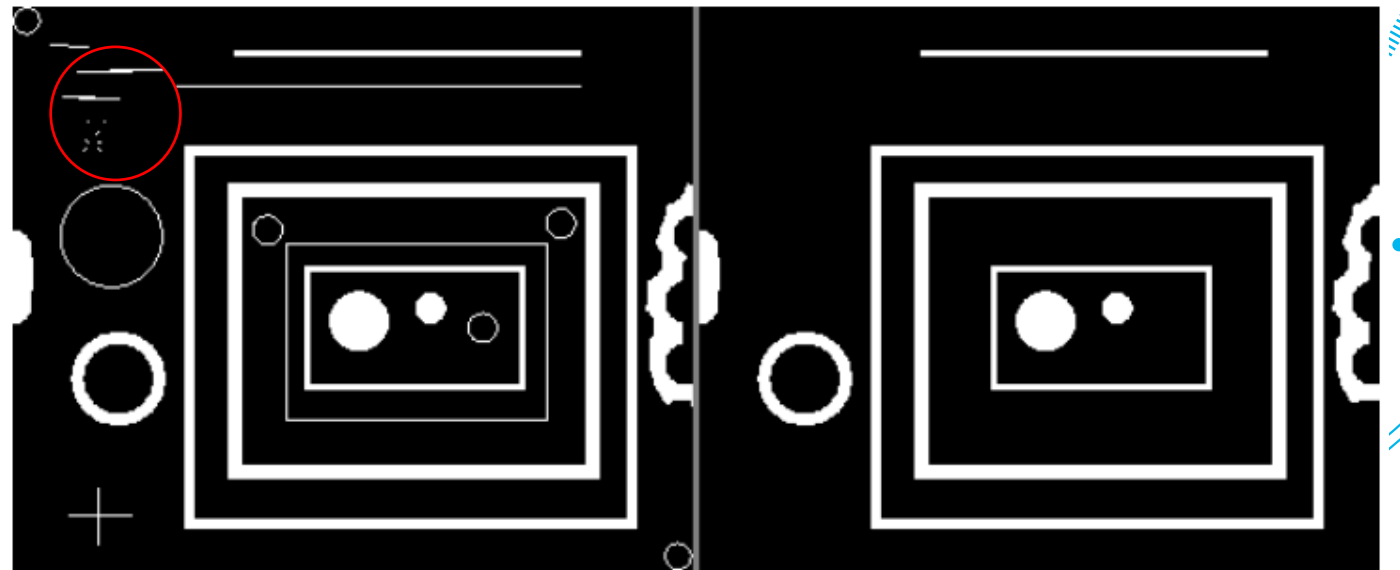
**Erosion**



# Mathematical Opening

The **opening** operation erodes an image and then dilates the eroded image, using the same structuring element for both operations.

This removes small objects and thin lines from an image, while preserving the shape and size of larger objects.

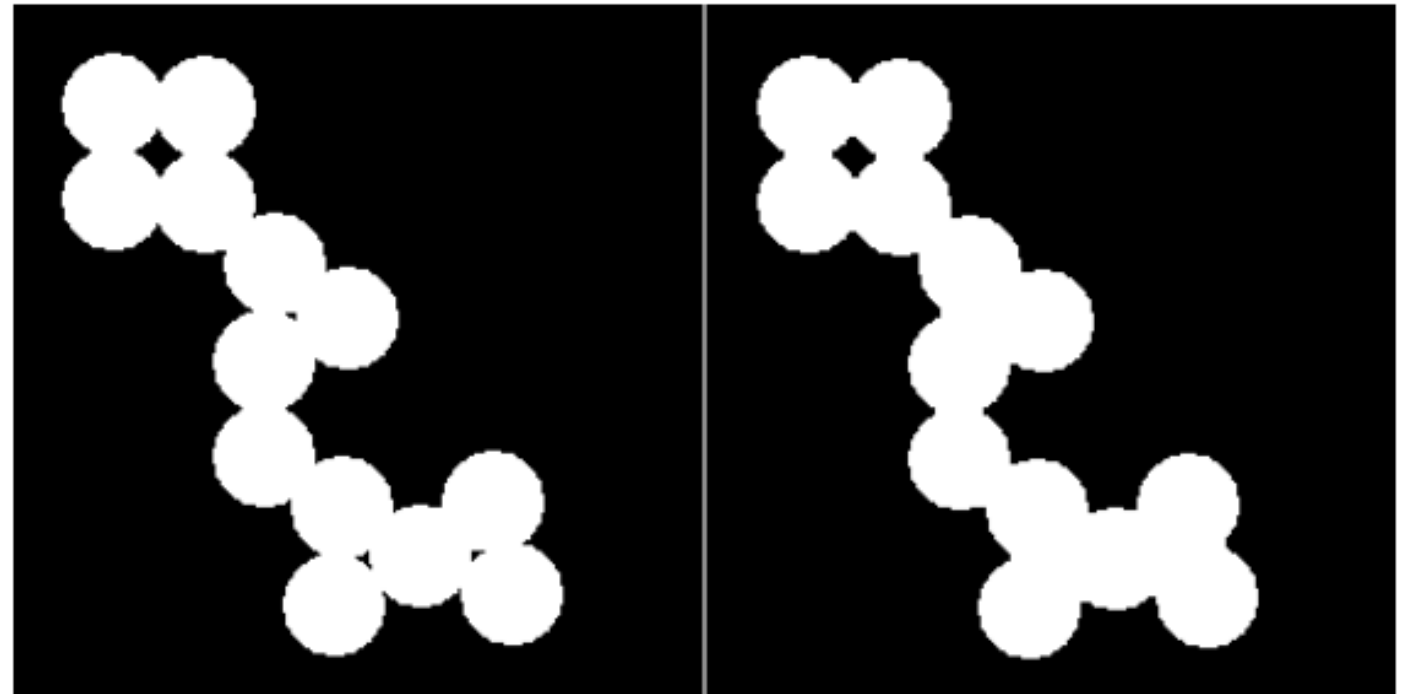




# Mathematical Closing

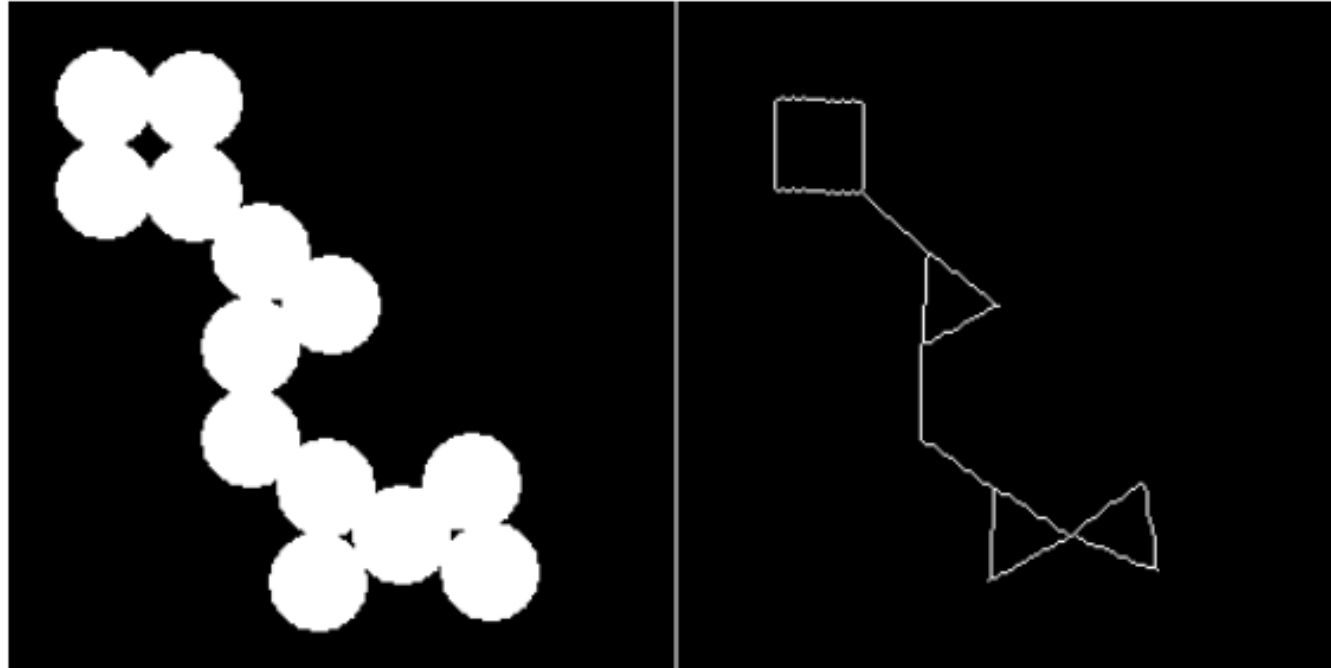
The **closing** operation dilates an image and then erodes the dilated image, using the same structuring element for both operations.

This results in filling small holes in an image, while preserving the shape and size of large holes and objects.



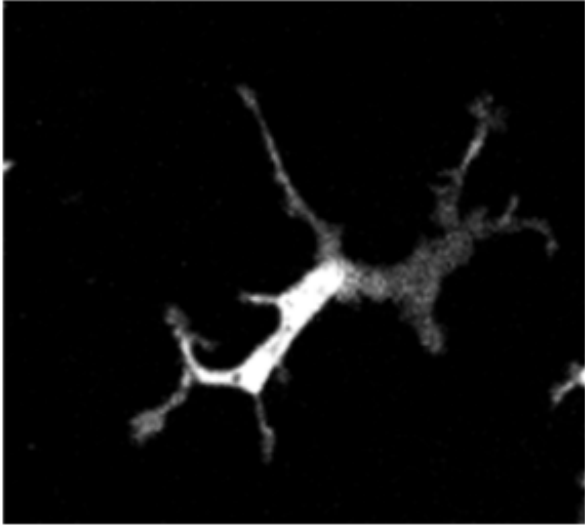
# Skeletonization

The process of **skeletonization** erodes all objects to centerlines without changing the essential structure of the objects, such as the existence of holes and branches.



# Skeletonization

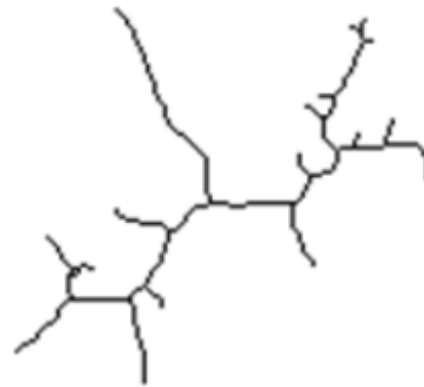
a



b



c



d

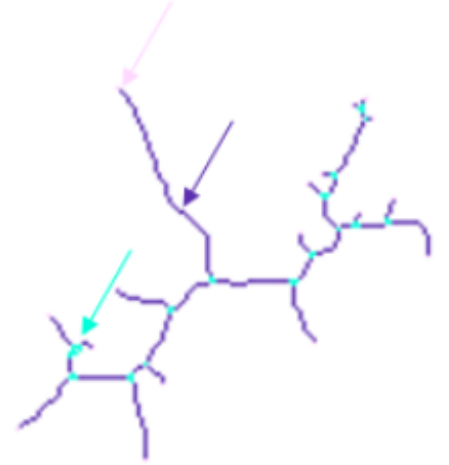


Image before and after skeletonization. a) raw image, b) binary image, c) skeleton image, d) tagged skeleton showing slab pixels (dark purple), junction pixels (cyan), and end-point pixels (pink). Examples of different skeleton pixels are indicated by arrows in the corresponding colors.